factors that influence immunogenicity can be divided under two categories.

1. Contribution of the immunogen to immunogenicity
2. Contribution of the biological system to immunogenicity

1. Contribution of the immunogen to immunogenicity

Immunogenicity is determined in part, by the following four properties of the immunogen.

A. Foreignness

The immune system normally discriminates between self and non self, so that only molecules that are foreign to the host are immunogenic. For example, albumin isolated from the serum of a rabbit and injected back into the same or another rabbit will not induce an immune response but the same protein when injected into other vertebrate species (rat) will induce an immune response.

B. Molecular size

There is a correlation between the size of a macromolecule and its immunogenicity. The best immunogens tend to have molecular mass approaching 100,000 daltons (Da). Generally, substances with a molecular mass less than 5000-10000 Da are poor immunogens; however a few substances with a molecular mass less than 1000 Da have proven to be immunogenic.

C. Chemical composition and complexity

Proteins are the most potent immunogens with polysaccharides ranking second. In contrast, lipids and nucleic acids of an infectious agent generally do not serve as immunogens unless they are complexed with proteins or polysaccharides (examples- lipoprotein or nucleo - protein). For example, attachment of tyrosine chains to the weakly immunogenic protein gelatin markedly enhances its immunogenicity.

D. Susceptibility to antigen processing and presentation

The development of both humoral and cell mediated immune responses requires interaction of T cells with antigen that has been processed and presented together with MHC (Major Histocompatibility Complex) molecules. To TH cells, the antigen must be presented with class II MHC molecules on an antigen presenting cell; to T_C Cells the antigen must be presented with class I MHC molecule on an altered self cell. 2. Contribution of the biological system to immunogenicity

Even if a macromolecule has the properties that contribute to immunogenicity, its ability to induce an immune response will depend on the following properties of the biological system that the antigen encounters.

A. Genetic constitution of the host animal

The genetic constitution (genotype) of an immunized animal plays an important role in determining whether a given substance will stimulate an immune response. Genetic control of immune responsiveness is largely made by genes mapping within the MHC

B. Immunogen dosage and route of administration

Whether an immunogen will induce an immune response also depends on the dose and mode of administration. A quantity of an immunogen that has no effect when injected intravenously may evoke a good antibody

TN_GOVT_XI_Micro_Biology_CH13.indd 237 16-02-2019 10.36.11 AM